

1. The company wants to automate its payroll process. Write a 'C' program to calculate the total salary by adding the basic salary and bonus amount of an employee.

A) PSEUDO CODE:-

START

READ basic_salary

READ bonus

total_salary = basic_salary + bonus

PRINT total_salary

STOP

INPUT:-

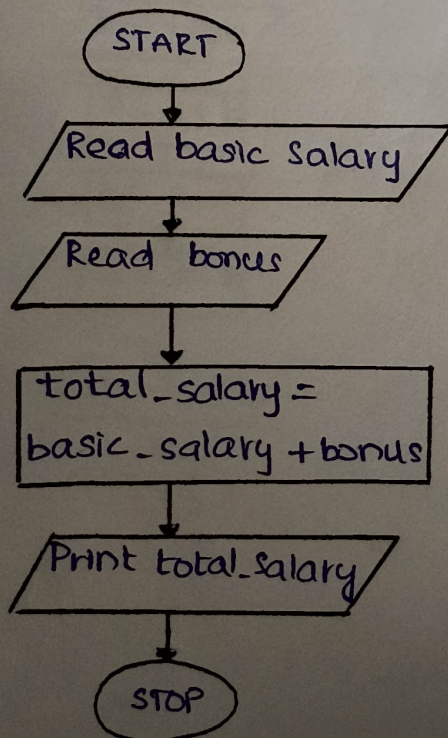
basic_salary = 30000

bonus = 5000

OUTPUT:-

Total Salary = 35000

FLOWCHART:-



2) During a database update, two employee IDs used needs to be exchanged. Write a c program to swap two numbers using a temporary variable.

A) PSEUDO CODE:-

START

READ a,b

temp = a

a = b

b = temp

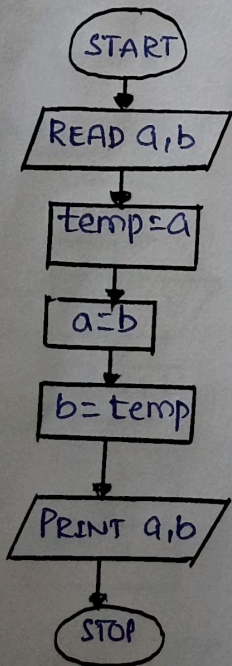
PRINT a,b

STOP

INPUT:- a = 5, b = 10

OUTPUT:- a = 10, b = 5

FLOW CHART:-



- 3) In an embedded system with limited memory, two values must be exchanged w/o using any extra storage. Write a c program to swap two numbers w/o using a temporary variable.

A) PSEUDO CODE:-

START

READ a,b

$a = a + b$

$b = a - b$

$a = a - b$

PRINT a,b

STOP

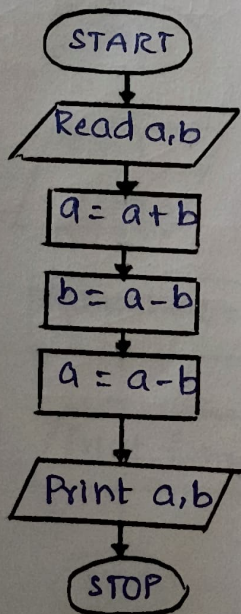
INPUT:-

$a = 5, b = 10$

OUTPUT:-

$a = 10, b = 5$

FLOWCHART:-



- 4) A Bank wants to provide customers with an estimate of the interest earned on their deposits. Write a C program to calculate the Simple Interest using the formula: $SI = (P \times R \times T) / 100$.

A) PSEUDO CODE:-

START

READ P, R, T

$SI = (P \times R \times T) / 100$

PRINT SI

STOP

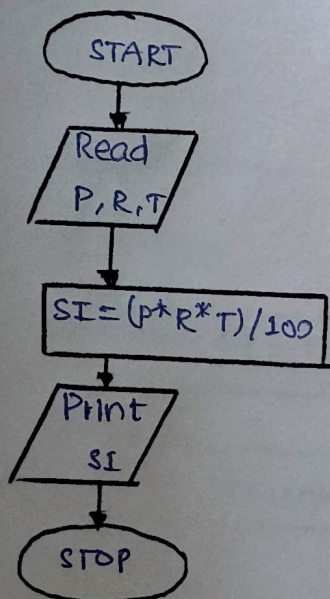
INPUT:-

P = 10000, R = 5, T = 2

OUTPUT:-

Simple Interest = 1000

FLOWCHART:-



- 5) A weather station records temperature in celsius, but reports need to be generated in fahrenheit. write a c program to convert temperature from celsius to Fahrenheit.

4) PSEUDO CODE:-

START

Read celsius

$\text{fahrenheit} = (\text{celsius} * 9/5) + 32$

PRINT fahrenheit

STOP

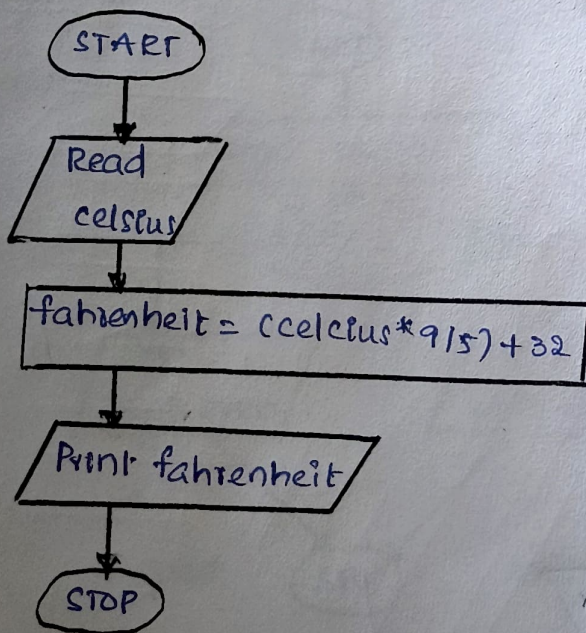
INPUT:-

celsius = 37

OUTPUT:-

Fahrenheit = 98.6

FLOW CHART:-



6) A mathematics learning application needs to be display both the quotient and remainder when two numbers are divided. Write a c program to find the quotient & remainder of two integers.

A) PSEUDO CODE:-

START

READ a, b

quotient = a/b

remainder = $a \% b$

PRINT quotient, remainder

STOP

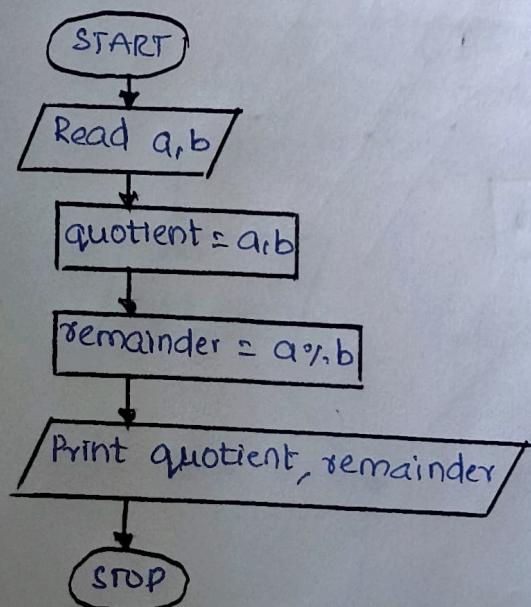
INPUT:-

$a = 17, b = 5$

OUTPUT:-

Quotient = 3, Remainder = 2

FLOW CHART:-



- 7) An event management system allows entry based on even numbered ticket IDs. Write a C program to determine whether a given ticket number is even or odd.

A) PSEUDO CODE:-

START

READ number

IF number $\% 2 == 0$ THEN

PRINT "Even"

ELSE

PRINT "Odd"

END IF

STOP

INPUT / OUTPUT:-

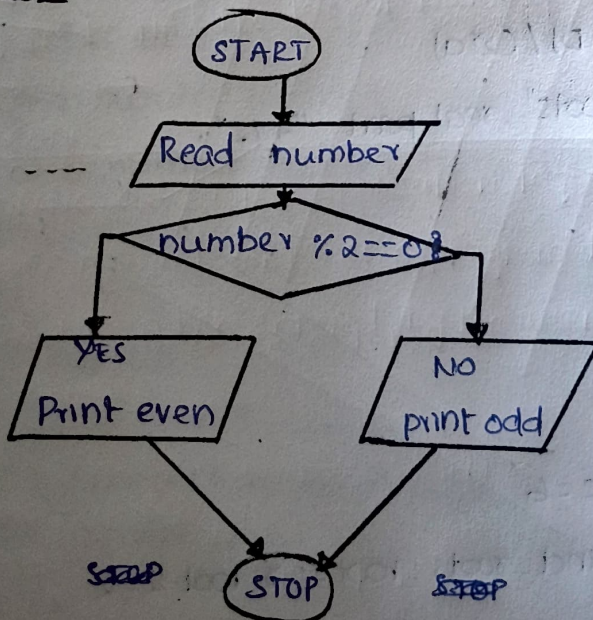
i/p number = 24

→ o/p = Even

i/p number = 17

→ o/p = odd

FLOW CHART:-



8) An engineer needs to solve quadratic equations of the form. Write a C program to calculate and display the roots of the quadratic equation.

A) PSEUDO CODE

START

READ a, b, c

$D = b^2 - 4ac$

IF $D > 0$ THEN

$root1 = (-b + \sqrt{D}) / (2a)$

$root2 = (-b - \sqrt{D}) / (2a)$

PRINT "Real and distinct roots", $root1, root2$

ELSE IF $D == 0$ THEN

$root1 = -b / (2a)$

PRINT "Real and equal roots", $root1$

ELSE

realPart = $-b / (2a)$

imagPart = $\sqrt{D} / (2a)$

PRINT "Complex roots", real part " $+/-$ ",

imagPart, " i "

END IF

STOP

INPUT / OUTPUT:-

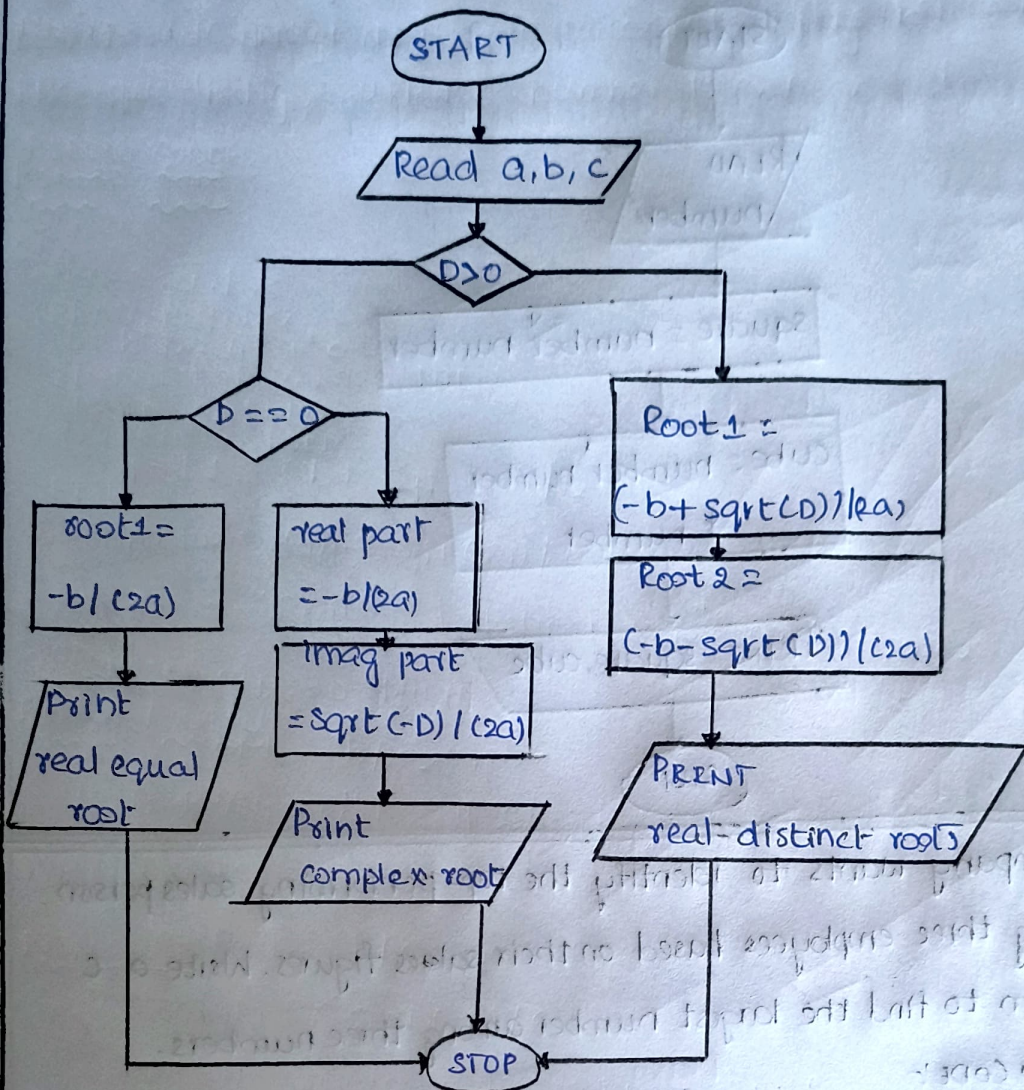
Input: $a=1, b=-5, c=6$

output: Real and distinct roots: $root1=3, root2=2$

Input: $a=1, b=2, c=5$

output: Complex roots: $-1.0 + 2.0i$ and $-1.0 - 2.0i$

FLOW CHART:-



- 9) A Scientific calculator application requires the square and cube of a given number. Write a c program to calculate the square & cube of a number entered by the user.

A) PSEUDO CODE:-

START

READ number

square = number * number

cube = number * number * number

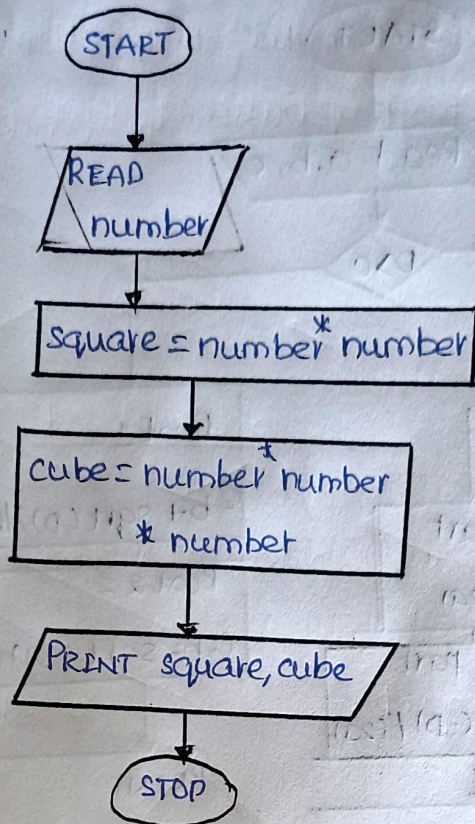
PRINT square, cube

STOP

INPUT / OUTPUT:-

number = 4 output: square = 16, cube = 64

FLOW CHART:-



- 10) A company wants to identify the top-performing sales person among three employees based on their sales figures. Write a C program to find the largest number among three numbers.

A) PSEUDO CODE:-

```
START
READ a, b, c
IF a >= b AND a >= c THEN
    largest = a
ELSE IF b >= a AND b >= c THEN
    largest = b
ELSE
    largest = c
ENDIF
PRINT Largest
STOP
```


INPUT/OUTPUT:-

i/p: $a=20, b=45, c=33$

o/p: Largest number = 45

FLOWCHART:-

